

Chapter 5 - Flow Matching and Similar Techniques

Exam Study Guide with Correctly Formatted Equations

Main exam goal: understand flow matching as probability-mass transport through a learned velocity field.

How to use this guide. First memorize the equation sheet, then understand the interpretation: velocity field = wind, ODE = particle motion, and the continuity equation = density transport. Finally, practice the sample questions and flashcards. Pay attention to differences between flow matching, diffusion, VAE, optimal-transport flow, latent flow, and consistency models.

1. Core Equation Sheet

Concept	Correct symbolic form
Transport ODE	$\frac{dx}{dt} = v_\theta(x, t), \quad t \in [0, 1]$
Base and target	$x(0) \sim p_0, \quad x(1) \sim p_1 = p_{\text{data}}$
Continuity equation	$\frac{\partial p_t(x)}{\partial t} + \nabla \cdot (p_t(x) v(x, t)) = 0$
Linear bridge	$x_t = (1 - t)x_0 + tx_1$
Velocity target	$u_t = \frac{dx_t}{dt} = x_1 - x_0$
Flow matching loss	$\mathcal{L}(\theta) = \mathbb{E}_{x_0, x_1, t} [\ v_\theta(x_t, t) - u_t\ _2^2]$
Euler update	$x_{n+1} = x_n + \Delta t v_\theta(x_n, t_n)$
Integral solution	$x(1) = x(0) + \int_0^1 v_\theta(x(t), t) dt$
Classifier-free guidance	$v_{\text{CFG}}(x, t, y) = v_\theta(x, t, \emptyset) + s[v_\theta(x, t, y) - v_\theta(x, t, \emptyset)]$
OT coupling	$\pi^* \in \arg \min_{\pi \in \Pi(p_0, p_1)} \mathbb{E}_{(x_0, x_1) \sim \pi} [\ x_0 - x_1\ _2^2]$
Consistency condition	$f_\theta(x_s, s) \approx f_\theta(x_t, t), \quad s < t$
Continuous diffusion SDE	$dx = f(x, t) dt + g(t) dW_t$

2. Main Exam Topics

2.1 Flow matching as transport

Flow matching treats generation as a deterministic transport problem: move samples from an easy base distribution p_0 , usually Gaussian noise, toward a complex target distribution $p_1 = p_{\text{data}}$. The model learns a time-dependent velocity field $v_\theta(x, t)$, which tells each particle where to move and how fast.

$$x(0) \sim p_0, \quad x(1) \sim p_1 = p_{\text{data}}$$

$$\frac{dx}{dt} = v_\theta(x, t), \quad t \in [0, 1]$$

Sampling is done by integrating this ODE from $t = 0$ to $t = 1$.

2.2 Continuity equation: distribution-level view

Flow matching is not only about individual particles. It also explains how the full probability density $p_t(x)$ evolves over time. The continuity equation states that probability mass is conserved: it moves through space, but it does not appear or disappear.

$$\frac{\partial p_t(x)}{\partial t} + \nabla \cdot (p_t(x)v(x, t)) = 0$$

For a vector field $F(x)$, divergence is

$$\nabla \cdot F(x) = \sum_{i=1}^d \frac{\partial F_i(x)}{\partial x_i}$$

Along a particle path $x(t)$, the log-density changes as

$$\frac{d}{dt} \log p_t(x(t)) = -\nabla \cdot v(x(t), t)$$

If the flow converges into a region, the density increases. If the flow diverges out of a region, the density decreases.

2.3 Bridge, coupling, and target velocity

To create supervised targets, flow matching pairs a base sample x_0 with a data sample x_1 . This pairing is called a coupling. A bridge defines intermediate states between them. The simplest bridge is linear interpolation.

$$x_t = (1 - t)x_0 + tx_1$$

For this bridge, the velocity target is known analytically:

$$u_t = \frac{dx_t}{dt} = x_1 - x_0$$

The model is then trained by direct vector-field regression:

$$\mathcal{L}(\theta) = \mathbb{E}_{x_0, x_1, t} \left[\|v_\theta(x_t, t) - u_t\|_2^2 \right]$$

This is why flow matching training is often easier to reason about than likelihood or score estimation.

2.4 Euler sampling and numerical ODE solving

After training, we sample by starting from $x_0 \sim p_0$ and numerically integrating the learned ODE until $t = 1$. Euler is the simplest solver. It follows the current velocity for a small step Δt .

$$x_{n+1} = x_n + \Delta t v_\theta(x_n, t_n)$$

The exact ODE solution can be written as an integral:

$$x(1) = x(0) + \int_0^1 v_\theta(x(t), t) dt$$

Euler approximates this integral by a Riemann sum. More accurate solvers such as Heun/RK2, RK4, or adaptive solvers can reduce integration error.

2.5 Why flow matching avoids VAE and diffusion bottlenecks

Flow matching does not require an approximate latent posterior as in a VAE:

$$q_\phi(z \mid x)$$

It also does not directly estimate the diffusion score:

$$\nabla_x \log p_t(x)$$

Instead, it learns a velocity field:

$$v_\theta(x, t)$$

The learning problem becomes supervised vector-field regression: match the network output to the bridge-defined target velocity.

2.6 Optimal-transport couplings

Naive flow matching may pair base and data samples independently. This can produce long or crossing trajectories. OT flow matching changes the coupling so paired samples are more coherent and the transport cost is reduced.

$$\pi^* \in \arg \min_{\pi \in \Pi(p_0, p_1)} \mathbb{E}_{(x_0, x_1) \sim \pi} [\|x_0 - x_1\|_2^2]$$

Even if the bridge remains linear, changing the coupling changes the geometry of the supervision.

2.7 Latent flow matching

Latent flow matching first encodes data into a latent space and learns the velocity field there. This can be easier than pixel-space transport because the latent space is lower-dimensional and often more semantic.

$$z = E_\psi(x), \quad \hat{x} = D_\psi(z)$$

$$\frac{dz}{dt} = v_\theta(z, t)$$

$$z_t = (1 - t)z_0 + tz_1, \quad u_t = z_1 - z_0$$

Sampling sequence: sample z_0 , integrate the latent ODE to obtain z_1 , then decode $\hat{x} = D_\psi(z_1)$.

2.8 Conditional flow matching and guidance

Conditional flow matching gives the velocity field access to a label or condition y :

$$\frac{dx}{dt} = v_\theta(x, t, y)$$

Classifier-free guidance trains the same model with and without the condition. During sampling, the conditional direction is amplified relative to the unconditional direction:

$$v_{\text{CFG}}(x, t, y) = v_\theta(x, t, \emptyset) + s[v_\theta(x, t, y) - v_\theta(x, t, \emptyset)]$$

Classifier-guided flow matching adds a classifier gradient:

$$\frac{dx}{dt} = v_\theta(x, t) + \lambda \nabla_x \log p_\phi(y \mid x, t)$$

Stronger guidance usually improves condition adherence but may reduce diversity.

2.9 Consistency models

Consistency models try to learn a map that sends different noisy versions of the same sample to the same canonical clean output.

$$f_\theta(x_s, s) \approx f_\theta(x_t, t), \quad s < t$$

A boundary condition anchors the model at clean data:

$$f_{\theta}(x_0, 0) = x_0$$

A common stop-gradient form is

$$\mathcal{L}_{\text{cons}} = \|f_{\theta}(x_s, s) - \text{sg}(f_{\theta}(x_t, t))\|_2^2$$

They are attractive because they can enable one-step or few-step generation, but they are fragile because consistency alone can admit trivial constant solutions.

2.10 Continuous-time diffusion connection

Discrete DDPM can be viewed as a finite-step discretization of a continuous-time stochastic differential equation:

$$dx = f(x, t) dt + g(t) dW_t$$

A common noisy interpolation is

$$x_t = \alpha(t)x_0 + \sigma(t)\varepsilon$$

For variance-preserving schedules,

$$\alpha(t)^2 + \sigma(t)^2 = 1, \quad \alpha(T) \approx 0$$

These ideas connect flow matching, diffusion, and consistency models under a broader transport view.

3. Sample Exam Questions and Answers

Question 1

Why is flow matching best understood as a transport model rather than simply a regression model?

The training loss is a regression loss,

$$\mathcal{L}(\theta) = \mathbb{E} \left[\|v_{\theta}(x_t, t) - u_t\|_2^2 \right]$$

but the learned object is a velocity field that induces the ODE

$$\frac{dx}{dt} = v_{\theta}(x, t)$$

This ODE transports the whole density from p_0 to p_1 . The continuity equation

$$\frac{\partial p_t}{\partial t} + \nabla \cdot (p_t v_t) = 0$$

shows that the deeper concept is distributional mass transport, not only pointwise MSE.

Question 2

What is the conceptual role of the interpolation path x_t , and why is it not a trivial design choice?

The interpolation path defines the intermediate states on which the network is trained. For the linear bridge,

$$x_t = (1 - t)x_0 + tx_1$$

the target is

$$u_t = x_1 - x_0$$

Different bridges $I(x_0, x_1, t)$ produce different intermediate distributions and target velocities:

$$u_t = \frac{\partial I(x_0, x_1, t)}{\partial t}$$

Therefore, the bridge controls how easy or difficult the learned vector field will be.

Question 3

Why can latent flow matching be preferable to pixel-space flow matching?

Pixel space is high-dimensional and contains local detail that can make transport difficult. Latent flow matching uses

$$z = E_\psi(x), \quad \frac{dz}{dt} = v_\theta(z, t)$$

This can be lower-dimensional and more semantically organized. It can reduce computation and simplify trajectories, but its quality depends on the encoder-decoder representation.

Question 4

Why do OT couplings matter even if the bridge formula remains linear?

With the same bridge

$$x_t = (1 - t)x_0 + tx_1$$

the pairing between x_0 and x_1 still determines the displacement $x_1 - x_0$. Random pairings may create long, crossing paths. OT couplings minimize expected transport cost:

$$\pi^* \in \arg \min_{\pi \in \Pi(p_0, p_1)} \mathbb{E}_{(x_0, x_1) \sim \pi} \left[\|x_0 - x_1\|_2^2 \right]$$

This produces shorter and more coherent directions that are easier for the velocity network to learn.

Question 5

Why are consistency models attractive, and why are they fragile?

They are attractive because they aim to map noisy states directly to a canonical clean output:

$$f_{\theta}(x_s, s) \approx f_{\theta}(x_t, t)$$

This can enable one-step or few-step generation. They are fragile because a trivial constant map can satisfy consistency without preserving data structure, so training often requires teacher distillation, boundary anchoring, or carefully designed objectives.

4. Flashcards - All Unique Questions

These cards keep the full unique review set while formatting formulas as proper symbolic equations.

Flashcard 1/50

Q: What is the role of the time-dependent velocity field $v_{\theta}(x, t)$ in flow matching?

A: It defines the direction and speed used to move particles from the base distribution p_0 to the target distribution p_1 .

Flashcard 2/50

Q: What is the ODE used for sampling in flow matching?

A:

$$\frac{dx}{dt} = v_{\theta}(x, t), \quad t \in [0, 1].$$

Flashcard 3/50

Q: In generative ODEs, what does the base distribution p_0 typically represent?

A: A simple distribution that is easy to sample from, usually Gaussian noise such as $\mathcal{N}(0, I)$.

Flashcard 4/50

Q: In flow matching, what does the target distribution p_1 represent?

A: The complex data distribution, usually $p_1 = p_{\text{data}}$.

Flashcard 5/50

Q: Write the continuity equation for the evolution of $p_t(x)$.

A:

$$\frac{\partial p_t(x)}{\partial t} + \nabla \cdot (p_t(x)v(x, t)) = 0.$$

Flashcard 6/50

Q: In the continuity equation, what does $\nabla \cdot (p_t(x)v(x, t))$ represent?

A: It is the divergence of probability flux; it measures the net outflow of probability mass from a region.

Flashcard 7/50

Q: What is the intuitive meaning of $\nabla \cdot F > 0$ at a point x ?

A: The vector field is expanding outward near x , creating net outflow from that point.

Flashcard 8/50

Q: Why does $p_t(x)$ increase where the flow converges, meaning $\nabla \cdot (p_t v) < 0$?

A: More probability mass enters the region than leaves it, so density accumulates.

Flashcard 9/50

Q: Along a particle trajectory $x(t)$, how does log-density change?

A:

$$\frac{d}{dt} \log p_t(x(t)) = -\nabla \cdot v(x(t), t).$$

Flashcard 10/50

Q: What is a coupling in flow matching?

A: A rule or joint distribution that pairs base samples $x_0 \sim p_0$ with data samples $x_1 \sim p_1$.

Flashcard 11/50

Q: Define x_t using the simplest linear interpolation bridge.

A:

$$x_t = (1 - t)x_0 + tx_1.$$

Flashcard 12/50

Q: What is the true velocity target u_t for the linear bridge $x_t = (1 - t)x_0 + tx_1$?

A:

$$u_t = \frac{dx_t}{dt} = x_1 - x_0.$$

Flashcard 13/50

Q: In the flow matching loss, what three components are sampled?

A: $x_0 \sim p_0$, $x_1 \sim p_1$, and $t \sim \text{Uniform}[0, 1]$.

Flashcard 14/50

Q: Write the standard flow matching regression loss.

A:

$$\mathcal{L}(\theta) = \mathbb{E}_{x_0, x_1, t} \left[\|v_\theta(x_t, t) - (x_1 - x_0)\|_2^2 \right].$$

Flashcard 15/50

Q: Why is flow matching considered supervised learning?

A: Because v_θ is directly regressed against known target velocities u_t defined by the chosen bridge.

Flashcard 16/50

Q: What is the simplest numerical method for integrating the flow matching ODE?

A: Forward Euler integration.

Flashcard 17/50

Q: Write one Euler update for flow matching sampling.

A:

$$x_{k+1} = x_k + \Delta t v_\theta(x_k, t_k).$$

Flashcard 18/50

Q: The Euler update is derived from which expansion?

A: The first-order Taylor expansion of $x(t + \Delta t)$.

Flashcard 19/50

Q: How does Euler approximate $x(1) = x(0) + \int_0^1 v_\theta(x(t), t) dt$?

A: It replaces the continuous integral by a Riemann sum of instantaneous velocities.

Flashcard 20/50

Q: In flow matching, what does $\|v_\theta(x, t)\|_2$ represent?

A: The instantaneous speed of the particle at position x and time t .

Flashcard 21/50

Q: How does flow matching avoid the score bottleneck in diffusion models?

A: It learns a deterministic velocity field by regression instead of estimating $\nabla_x \log p_t(x)$.

Flashcard 22/50

Q: What is the main sampling difference between flow matching and standard diffusion?

A: Flow matching uses deterministic ODE integration, while standard diffusion often uses stochastic reverse SDE or Markov sampling.

Flashcard 23/50

Q: Why use optimal-transport couplings in flow matching?

A: To pair x_0 and x_1 so that transport cost is minimized, producing straighter and easier-to-learn trajectories.

Flashcard 24/50

Q: What cost is typically minimized by an OT coupling π between p_0 and p_1 ?

A:

$$\pi^* \in \arg \min_{\pi \in \Pi(p_0, p_1)} \mathbb{E}_{(x_0, x_1) \sim \pi} [\|x_0 - x_1\|_2^2].$$

Flashcard 25/50

Q: How does OT improve the geometry of the learned velocity field?

A: It reduces path crossing and creates more coherent displacement directions by pairing base and target samples more sensibly.

Flashcard 26/50

Q: What are the two learned modules in latent flow matching?

A: An encoder-decoder representation model, such as AE/VAE, and a latent velocity model $v_\theta(z, t)$.

Flashcard 27/50

Q: What is the sampling sequence for latent flow matching?

A: Sample $z_0 \sim \mathcal{N}(0, I)$, integrate $dz/dt = v_\theta(z, t)$ to get z_1 , then decode $\hat{x} = D_\psi(z_1)$.

Flashcard 28/50

Q: What is the CFG formula for the guided velocity in flow matching?

A:

$$v_{\text{CFG}} = v_{\text{uncond}} + s(v_{\text{cond}} - v_{\text{uncond}}).$$

Flashcard 29/50

Q: What does the guidance scale s represent in CFG?

A: It controls the strength of the condition-specific push relative to the generic data-manifold direction.

Flashcard 30/50

Q: How is v_{uncond} obtained in a model trained with label dropout?

A: By querying the model with the null/no-label condition $\emptyset$.

Flashcard 31/50

Q: In classifier-guided flow matching, how is the classifier gradient incorporated?

A:

$$\frac{dx}{dt} = v_{\theta}(x, t) + \lambda \nabla_x \log p_{\phi}(y \mid x, t).$$

Flashcard 32/50

Q: What consistency condition defines a consistency model $f_{\theta}(x, t)$?

A:

$$f_{\theta}(x_s, s) \approx f_{\theta}(x_t, t), \quad s < t.$$

Flashcard 33/50

Q: What is the mapping target of a consistency model $f_{\theta}(x, t)$?

A: A canonical clean representation of the sample, usually x_0 .

Flashcard 34/50

Q: How do consistency models achieve one-step generation?

A: They learn a direct map from a noisy state x_t to the clean output x_0 in one function evaluation.

Flashcard 35/50

Q: In consistency training, what is the purpose of the boundary loss at $t = 0$?

A: It anchors the model so that $f_{\theta}(x_0, 0) = x_0$, making it act like the identity at zero noise.

Flashcard 36/50

Q: What is consistency distillation?

A: Training a consistency student model to match outputs or trajectories produced by a stronger pre-trained teacher model.

Flashcard 37/50

Q: Write the general form of a continuous-time diffusion SDE.

A:

$$dx = f(x, t) dt + g(t) dW_t.$$

Flashcard 38/50

Q: What defines the variance-preserving (VP) property?

A: The signal scale $\alpha(t)$ and noise scale $\sigma(t)$ satisfy

$$\alpha(t)^2 + \sigma(t)^2 = 1.$$

Flashcard 39/50

Q: For $x_t = \alpha(t)x_0 + \sigma(t)\varepsilon$, what is the boundary condition for $\alpha(t)$ as $t \rightarrow T$?

A: $\alpha(T) \approx 0$, meaning the original signal vanishes into noise.

Flashcard 40/50

Q: How is the student updated in stop-gradient consistency training?

A:

$$\mathcal{L}_{\text{cons}} = \|f_{\theta}(x_s, s) - \text{sg}(f_{\theta}(x_t, t))\|_2^2.$$

Flashcard 41/50

Q: In 2D image flow matching, what does a pixel (i, j) in the velocity map represent?

A: The instantaneous rate of change of that pixel brightness at time t .

Flashcard 42/50

Q: Why is per-pixel velocity $[v_{\theta}(x, t)]_{ij}$ not independent of other pixels?

A: The neural network uses the whole image context to decide the correct change for each pixel.

Flashcard 43/50

Q: What is the main lever changed in optimal transport flow compared with baseline flow matching?

A: The coupling between x_0 and x_1 is changed to minimize transport cost, while the bridge may remain linear.

Flashcard 44/50

Q: What happens when the bridge $I(x_0, x_1, t)$ is changed?

A: It changes the intermediate states x_t and the target velocity

$$u_t = \frac{\partial I(x_0, x_1, t)}{\partial t}.$$

Flashcard 45/50

Q: What is the relationship between discrete DDPM and continuous-time SDEs?

A: DDPM can be viewed as a finite-step discretization of a continuous-time diffusion SDE.

Flashcard 46/50

Q: In teacher-guided distillation, what does $p_{\theta}(x_0 | x_t, t)$ represent?

A: The teacher model's estimate of the clean data x_0 given the noisy observation x_t .

Flashcard 47/50

Q: How is a teacher denoise step from t to $s < t$ implemented using Monte Carlo sampling?

A: Sample a clean candidate x_0 from the teacher posterior, then re-noise it to time s using $q(x_s | x_0, s)$.

Flashcard 48/50

Q: What is the benefit of sampling in log-SNR space during consistency training?

A: It avoids oversampling extremely noisy regions where the signal is too weak to provide useful gradients.

Flashcard 49/50

Q: What does a negative velocity value $[v_\theta]_{ij} < 0$ imply for a pixel normalized to $[-1, 1]$?

A: The pixel should become darker, moving toward -1 .

Flashcard 50/50

Q: What is the deterministic transport view of flow matching?

A: The velocity field defines a flow map that pushes the entire probability mass from p_0 to p_1 through deterministic ODE paths.

5. One-Page Final Review Checklist

- Can you explain why $v_\theta(x, t)$ is like a wind field?
- Can you write $\frac{dx}{dt} = v_\theta(x, t)$ and explain sampling from p_0 to p_1 ?
- Can you write and interpret $\frac{\partial p_t}{\partial t} + \nabla \cdot (p_t v) = 0$?
- Can you derive $u_t = x_1 - x_0$ from $x_t = (1 - t)x_0 + tx_1$?
- Can you explain why $\mathcal{L}(\theta) = \mathbb{E}[\|v_\theta(x_t, t) - u_t\|_2^2]$ is supervised regression?
- Can you write Euler integration and explain why it is a Riemann-sum approximation?
- Can you compare flow matching with VAE and diffusion?
- Can you explain why OT couplings reduce path crossing?
- Can you explain latent flow matching and its sampling sequence?
- Can you write $v_{\text{CFG}} = v_{\text{uncond}} + s(v_{\text{cond}} - v_{\text{uncond}})$?
- Can you explain why consistency models are fast but fragile?

Most important chain to memorize:

$$x_0 \sim p_0, \quad x_1 \sim p_1, \quad t \sim U[0, 1], \quad x_t = (1 - t)x_0 + tx_1,$$

$$u_t = x_1 - x_0, \quad \mathcal{L}(\theta) = \mathbb{E}[\|v_\theta(x_t, t) - u_t\|_2^2].$$

End of Chapter 5 study guide.